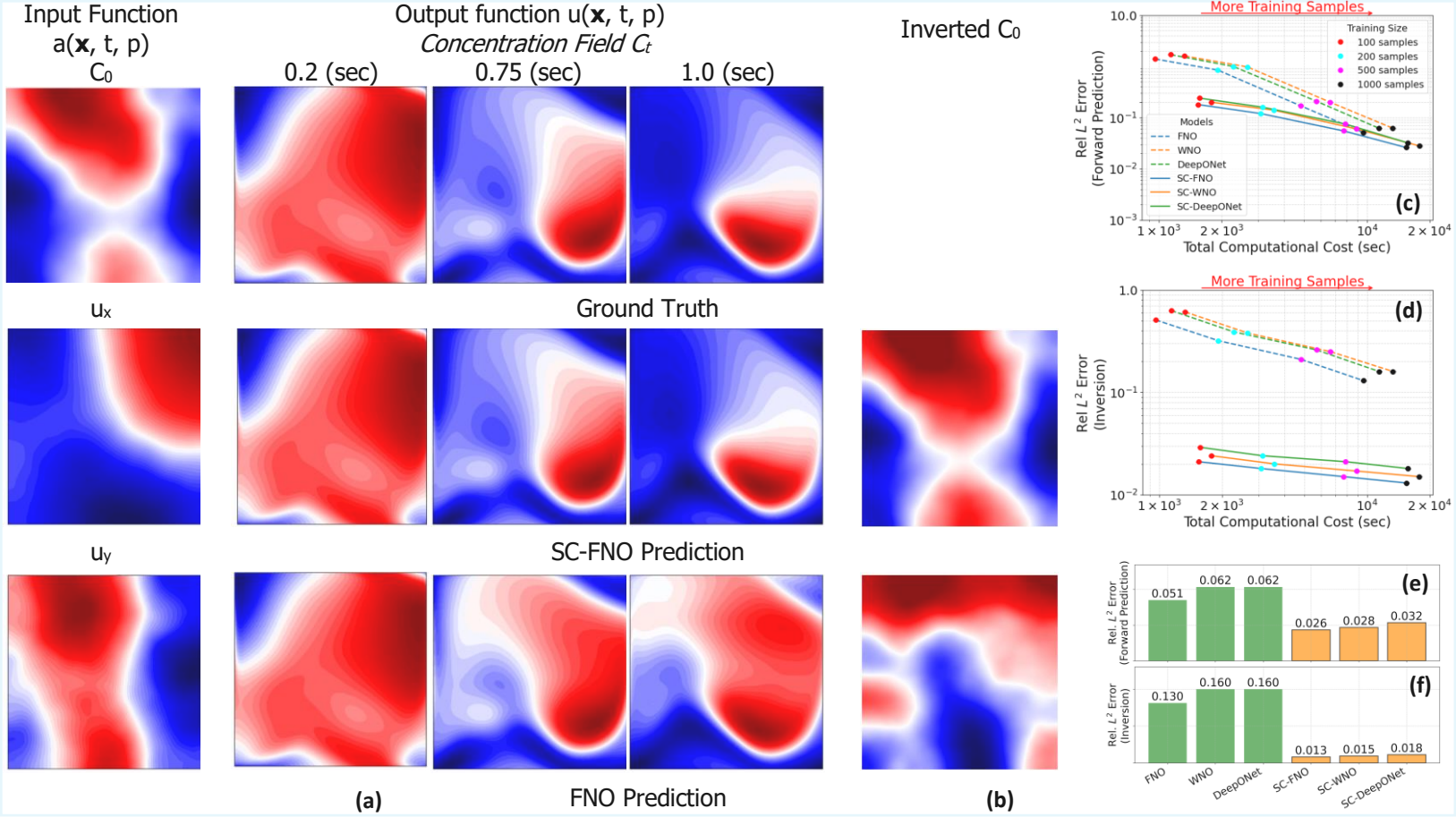
Input Function
a(x, t, p)
C0
Output function u(x, t, p)
Concentration Field Ct
0.2 (sec)
0.75 (sec)
1.0 (sec)
Inverted C0
ux
Ground Truth
uy
SC-FNO Prediction
FNO Prediction
(a)
(b)
More Training Samples
Training Size
100 samples
200 samples
500 samples
1000 samples
Models
FNO
WNO
DeepONet
SC-FNO
SC-WNO
SC-DeepONet
Rel L2 Error (Forward Prediction)
Total Computational Cost (sec)
(c)
Rel L2 Error (Inversion)
(d)
(e)
(f)
0.051
0.062
0.062
0.026
0.028
0.032
0.130
0.160
0.160
0.013
0.015
0.018